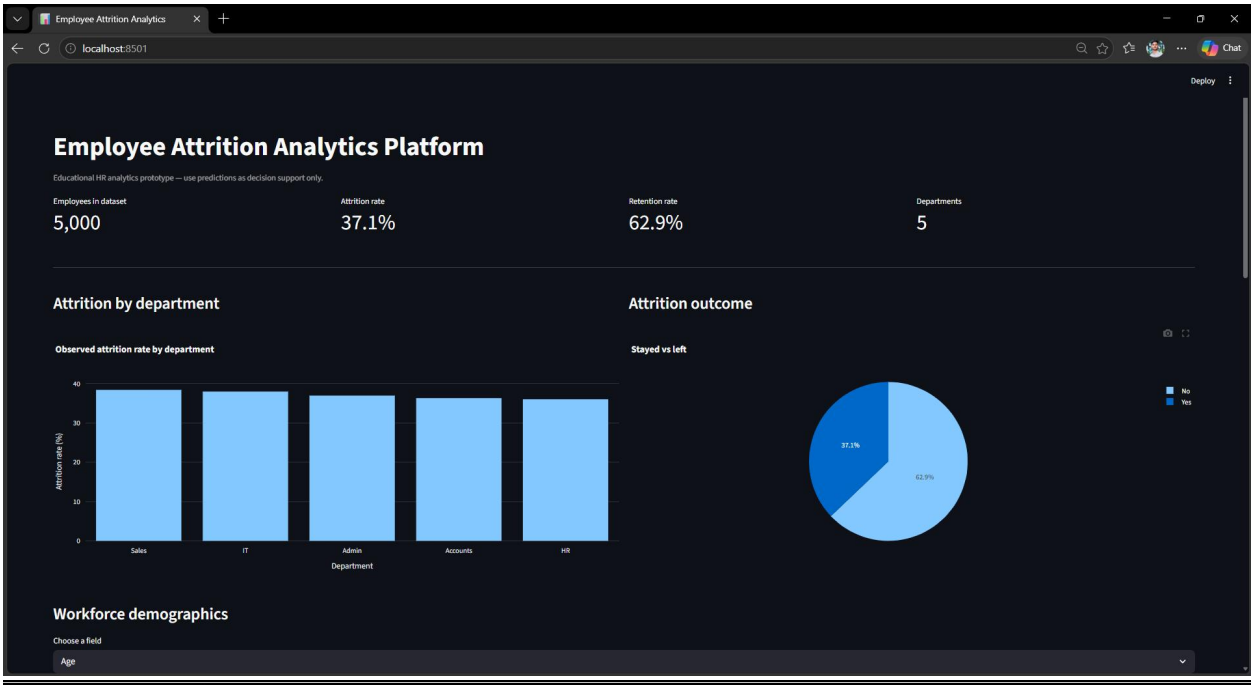
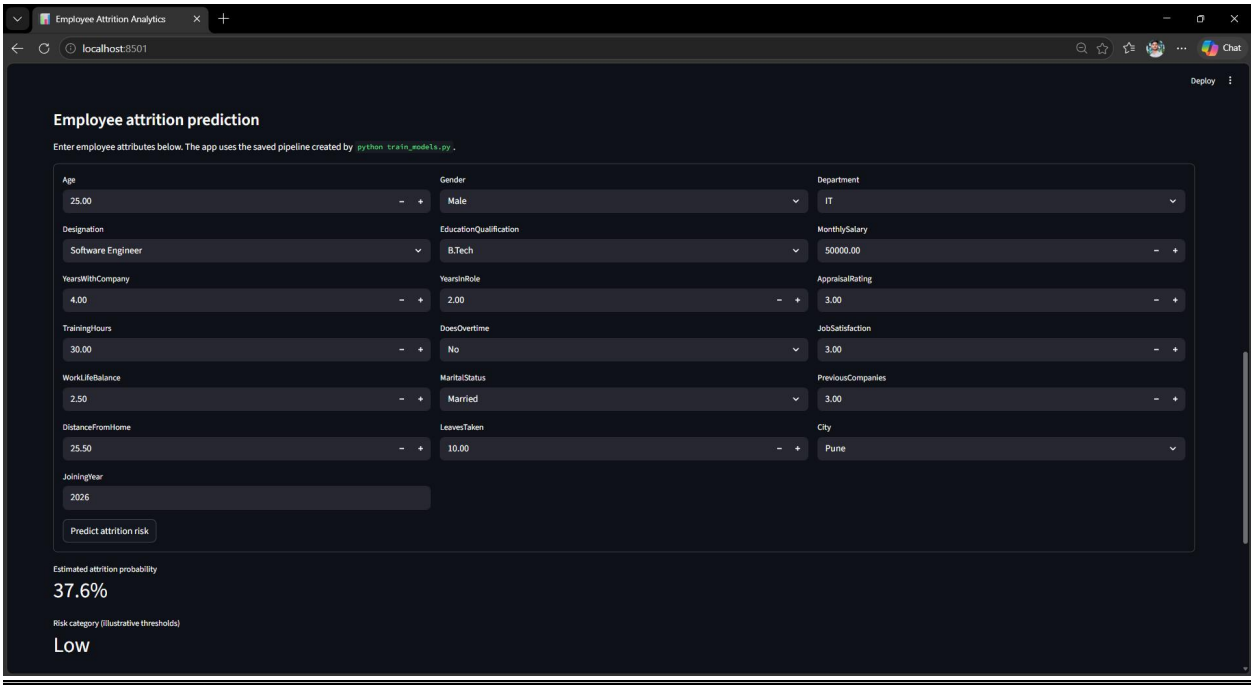
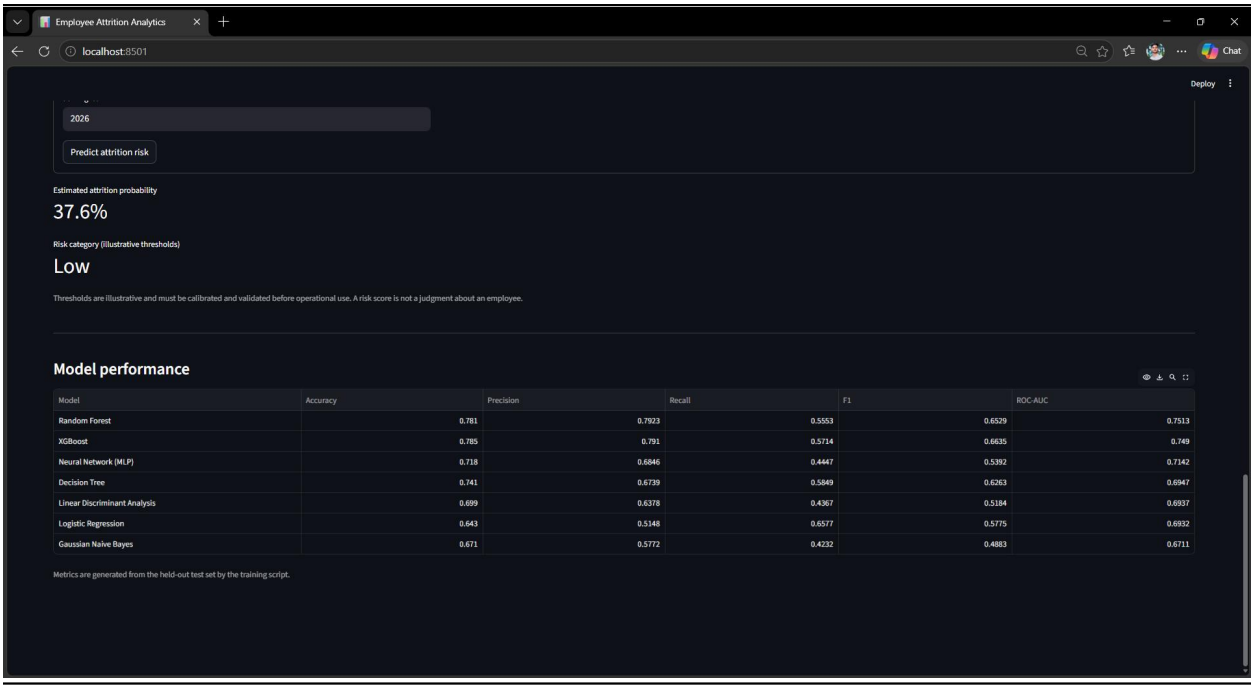

Employee Attrition Analytics
localhost:8501
Chat
Deploy
Employee Attrition Analytics Platform
Educational HR analytics prototype — use predictions as decision support only.
Employees in dataset
5,000
Attrition rate
37.1%
Retention rate
62.9%
Departments
5
Attrition by department
Observed attrition rate by department
Attrition rate (%)
Sales
IT
Admin
Accounts
HR
Department
Attrition outcome
Stayed vs left
No
Yes
37.1%
62.9%
Workforce demographics
Choose a field
Age


Employee Attrition Analytics
localhost:8501
Chat
Deploy
Sales
IT
Admin
Accounts
HR
Department
Workforce demographics
Choose a field
Age
Employee distribution by Age
Employees
Age
Employee attrition prediction
Enter employee attributes below. The app uses the saved pipeline created by python train_models.py .
Age
25.00
Gender
Male
Department
IT


Employee Attrition Analytics
localhost:8501
Chat
Deploy
Employee attrition prediction
Enter employee attributes below. The app uses the saved pipeline created by python train_models.py.
Age
25.00
Gender
Male
Department
IT
Designation
Software Engineer
EducationQualification
B.Tech
MonthlySalary
50000.00
YearsWithCompany
4.00
YearsInRole
2.00
AppraisalRating
3.00
TrainingHours
30.00
DoesOvertime
No
JobSatisfaction
3.00
WorkLifeBalance
2.50
MaritalStatus
Married
PreviousCompanies
3.00
DistanceFromHome
25.50
LeavesTaken
10.00
City
Pune
JoiningYear
2026
Predict attrition risk
Estimated attrition probability
37.6%
Risk category (illustrative thresholds)
Low


Employee Attrition Analytics
localhost:8501
Chat
Deploy
2026
Predict attrition risk
Estimated attrition probability
37.6%
Risk category (illustrative thresholds)
Low
Thresholds are illustrative and must be calibrated and validated before operational use. A risk score is not a judgment about an employee.
Model performance
Model
Accuracy
Precision
Recall
F1
ROC-AUC
Random Forest
0.781
0.7923
0.5553
0.6529
0.7513
XGBoost
0.785
0.791
0.5714
0.6635
0.749
Neural Network (MLP)
0.718
0.6846
0.4447
0.5392
0.7142
Decision Tree
0.741
0.6739
0.5849
0.6263
0.6947
Linear Discriminant Analysis
0.699
0.6378
0.4367
0.5184
0.6937
Logistic Regression
0.643
0.5148
0.6577
0.5775
0.6932
Gaussian Naive Bayes
0.671
0.5772
0.4232
0.4883
0.6711
Metrics are generated from the held-out test set by the training script.